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Tsi12C

Tenkasi District

Common Half Yearly Examination - December 2024

20-12-24

Standard 12

CHEMISTRY

Time Allowed: 3.00 Hours Maximum Marks: 70

PART - A

I. Choose the best answer: **15×1=15**

1) Which of the following plot gives Ellingham diagram?
   a) ΔSVST b) ΔG°VST c) ΔG°VS1/T d) ΔG°VST^2

2) The compound that is used in nuclear reactors as protetive shields and control rods is
   a) Metal Borides b) Metal Oxides c) Metal Carbonates d) Metal Carbide

3) Among the following the correct order of acidity is
   a) $HClO_2 < HClO < HClO_3 < HClO_4$ b) $HClO_4 < HClO_2 < HClO < HClO_3$
   c) $HClO_3 < HClO_4 < HClO_2 < HClO$ d) $HClO < HClO_2 < HClO_3 < HClO_4$

4) Permangante ion changes to _____ in acidic medium.
   a) MnO_4^{2-} b) Mn^{2+} c) Mn^{3+} d) MnO_2

5) A complex in which the oxidation number of the metal is zero is
   a) $K_4[Fe(CN)_6]$ b) $[Fe(Cr)_3(NH_3)_3]$ c) $[Fe(CO)_5$ d) Both (b) and (c)

6) Potassium has a bcc structure with nearest neighbour distance 4.524°. It's atomic weigh 39. It density will be
   a) 915 Kg m^{-3} b) 2142 Kg m^{-3} c) 452 Kg m^{-3} d) 390 Kg m^{-3}

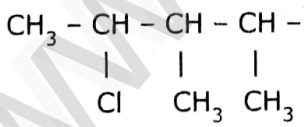
7) Time required for 100% completetion of a zero order reaction is
   a) $a/2K$ b) aK c) $2K/a$ d) a/K

8) P^H of a saturated solution of $Ca(OH)_2$ is 9. The solubility product of (Kcp) $CO(OH)_2$.
   a) 0.5×10^{-15} b) 0.25×10^{-10} c) 0.125×10^{-15} d) 0.5×10^{-10}

9) How many Faraday of electricity are required for the following reaction to occur $MnO_4^- \rightarrow Mn^{2+}$?
   a) 5F b) 3F c) 1F d) 7F

10) Hair cream is
   a) gel b) emulsion c) solid sol d) sol

11) The correct IUPAC name of the following compound.



   a) 4-chloro-2, 3-dimethyl pentan-1-ol b) 2, 3 dimethyl-4-chloro pentan-1-ol
   c) 2, 3, 4-trimethyl-4-chloro butan-1-ol d) 4 chloro-2, 3, 4-trimethyl pentan-1-ol

12) In the following reaction $CH \equiv CH \xrightarrow[HgSO_4]{H_2SO_4}$ X, product 'X' will not give
   a) Tollen's test b) Victor Mayer's test
   c) Iodoform test d) Fehling's solution test

13) Which one of the following will not undergo hoffman's reaction?
   a) $CH_3CONHCH_3$ b) $CH_3CH_2CONH_2$ c) CH_3CONH_2 d) $C_6H_5CONH_2$

14) The number of sp^2 and sp^3 hybridised carbon in fructose are
   a) 1 and 4 b) 4 and 2 c) 5 and 1 d) 1 and 5

15) As per BIS standard Grade 1 soap have _____ minimum TFM value.
   a) 70% b) 60% c) 67% d) 76%

Kindly Send Me Your Key Answer to Our email id - Padasalai.net@gmail.com

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PART - B

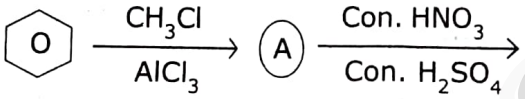
Answer any SIX questions: [Q.No. 23 is compulsory] **6×2=12**

16) What are the various steps involved in the extraction of pure metals from their ores?
17) Draw and explain the structure of Ammonia.
18) Explain Hume - Rothery rule.
19) What is linkage isomerism?
20) Explain Arhenius Equation.
21) What is common ion effect? Explain.
22) What is Tyndal effect?
23) How will you convert chloro benzene to phenol?
24) How will you prepare Nylon 66?

PART - C

Answer any SIX questions: [Q.No. 31 is compulsory] **6×3=18**

25) Write the uses of Silicon.
26) $[Ti(H_2O)_6]^{3+}$ is coloured where as $[Se(H_2O)_6]^{3+}$ is colourless. Why?
27) Give the difference between order and molecularity.
28) What are catalytic poisons? Give two examples.
29) Calculate the packing efficiency of simple cubic system.
30) Explain intermediate compound formation theory.
31) Identify A, B and C.



32) How will you prepare Terlene?
33) What are hormones? Give examples.

PART - D

Answer all the questions: **5×5=25**

34) a) i) Describe the method for refining nickel.
ii) What is autoreduction?

(OR)

b) i) **Complete the following:**

a) $H_2B_4O_7 \xrightarrow{\text{Red hot}}$

b) $B(OH)_3 + NH_3 \longrightarrow$

ii) Write note on Holme's signal.

35) a) Compare Lanthanides and Actinides.

(OR)

b) Explain Frenkal and Schotty defect.

36) a) What are the posulates of V.B. Theory?

(OR)

b) Derive Nernst Equation.

37) a) i) Write the difference between DNA and RNA.
ii) What is Zwitter Ion? Explain.

(OR)

b) Explain the mechanism of Cannizaro's Reaction.

38) a) Discuss the structure of glucose.

(OR)

b) i) How do antiseptic differ from disinfectants?
ii) What are food perservatives?

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